

NFL Scoring Trends: Strategy vs Pace

A DS 4002 Case Study



You have just been hired as a data analyst for an NFL front office. Teams across the league are constantly trying to improve offensive performance and increase scoring. Over the past decade, offenses have become more pass-heavy and games have changed in pace, leading many to believe that these factors directly increase scoring [2].

Your job is to investigate whether these assumptions are actually true.

Using real NFL play-by-play data from the 2010–2024 seasons [1], you will analyze how offensive strategy (pass rate) and game pace (plays per game) have changed over time and how they relate to scoring outcomes. You will explore trends in the data, build statistical models, and evaluate which factors are the strongest predictors of scoring.

Your final deliverable will be a data-driven analysis that explains how offensive strategy and pace impact scoring in the NFL and provides insight into what teams should prioritize moving forward.

[1] nflfastR, “nflfastR — NFL Play-by-Play Data for R and Analysis.” [Online]. Available: <https://nflfastR.com>. [Accessed: May 2026].

[2] J. R., “Analyzing NFL Offenses (1985–2022) – Passing Attempts,” Footballguys.com, Jul. 21, 2023. [Online]. Available: <https://www.footballguys.com/article/2023-analyzing-passing-attempts-dating-back-to-1985>. [Accessed: May 2026].